Cubic Time Dimensions in Quantum Mechanics, Finding Electron Orbital States with Time

Second edition: proper time, coordinate time, and the periodic table

Anthony M. Vargas Lyra R. Novak Aethus Samir-Iqbal

First edition October 12, 2024 · Second edition September 24, 2026

Abstract

This paper models atomic structure with time, not space, as the primary variable. This second edition keeps the first edition's premise, that an electron's states can be found from temporal interference, and replaces its fitted three-time model with a construction that has no per-atom parameters. Two clocks are used. The electron's own clock is its Compton phase, and a stationary state's clock is its energy phase. The Dirac virial theorem gives $E = mc^2\langle\beta\rangle$ exactly for inverse-square forces, and only for them. Reading $\langle\beta\rangle$ as the electron's average clock rate gives the premise, $E = mc^2\langle d\tau/dt\rangle$.

- *One-electron atoms.* The construction reproduces the reference ionization energies of H and He^+, and Herzberg's deuterium Lyman-α line, up to the Lamb shift. The residuals are $+2.48$, $+8.18$ and -3.26 ppm, each within 0.08 ppm of the measured or QED-scaled Lamb shift.
- *X-ray lines.* With the historical screening constants, the Moseley RMS error falls from 5.7% to 1.05%. The Kα doublet, which is exactly zero without relativity, comes out within 0.66% for $Z \geq 40$.
- *All 118 elements.* In any mean-field atom whose interactions are all inverse-square, the total energy is the sum of the electrons' proper-time deficits plus a nuclear-volume term of at most 0.6%. The identity closes to 1.1×10^{-7} in self-consistent Dirac–Slater atoms for all 118 elements. Each orbital energy splits into a *clock* term, a *shelter* term from the charge lying farther out, and an exchange term. With no per-atom parameters, first ionization energies agree with the 99 NIST reference values to 5.0% RMS (median 3.2%), and K-shell binding energies from neon to uranium agree to 0.46%.
- *Decay clocks.* Spontaneous emission obeys $\tau = 3T^3/(32\pi^3\alpha\, t_d^2)$, a cube of the transition period. It gives hydrogen's 2p lifetime to 0.1% and the alkali resonance lifetimes to within about 8%. Alpha decay follows the Sommerfeld ratio $\eta = Z_1 Z_2 \alpha/\beta$. With the textbook nuclear radius and no fitted constant, 35 half-lives spanning 29.8 decades are reproduced to 0.56 decades RMS. The scatter about a single offset depends only weakly on the radius (0.37–0.46 decades); the overall level does not.

The first edition's starting time t_1 turns out to be the electron's Compton clock.

Contents

1 Introduction **3**

2 What the first edition established, and what it left open **3**

3 Methodology: two clocks, one phase **4**
3.1 The premise, and what makes it exact . . . 4
3.2 The quantization, and every level as a clock rate . . . 5
3.3 The first edition's interference, with physical frequencies . . . 5
3.4 Many electrons . . . 5
3.5 Decay clocks . . . 7
3.6 Error metric . . . 7

4 Results: one-electron atoms **8**
4.1 Hydrogen, deuterium, He^{+} and U^{91+} . . . 8
4.2 X-ray lines of real atoms . . . 9

5 Results: all 118 elements **9**
5.1 The whole-atom ledger . . . 9
5.2 Ionization and K-shell binding . . . 10
5.3 What the ledger shows . . . 10

6 Results: the decay clocks **12**
6.1 Spontaneous emission and the cube of the period . . . 12
6.2 Alpha decay . . . 12
6.3 The radioactive edge of the table . . . 13

7 The first edition's fitted times on the clock ladder **13**

8 Spin in the cubic-time framework (retained, with corrections) **14**

9 Discussion **14**

10 Conclusion **15**

A Numerical methods and checks **15**

B Data sources **16**

C The periodic table in proper time **16**

D Alpha emitters **19**

E The radioactive edge: longest-lived isotopes **20**

Provenance tags. Each tag marks the status of the claim it follows.

- [FORCED-BY-DEFINITION]: follows from a definition or identification.
- [FORCED-BY-MECHANICS]: follows from the stated equations of motion with no further choice.

- [MEASURED]: a literature value, used only for grading and never fed back into a calculation.
- [ASSUMED]: a stated modelling choice.
- [FITTED]: adjusted to data.
- [OPEN]: a question raised and not settled.

The paper refers to equations of the first edition as "1st-ed. eq. (n)".

1 Introduction

Quantum mechanics has traditionally been built on spatial coordinates and wavefunctions. It gives a robust account of particles at the microscopic scale, but it offers little temporal labelling of the atom. Hydrogen is rarely written in seconds per radian, and we are not aware of a standard temporal description of the periodic table.

The first edition started from a practical question: could a reductive model with one spatial dimension x and several time dimensions pin down an electron's actual trajectory, bypassing the uncertainty of purely spatial descriptions? The idea came from earlier work in harmonic music theory in the Reaper DAW. Custom templates synchronized frequency-wave patterns to a master BPM clock and tracked them with high-resolution spectroscopy, Blackman–Harris FFT analysis, and a Lorenz attractor used to plot the pulses. These experiments showed "orbital" jumps in interference patterns that stabilized without collapsing, and that stayed stable as long as frequency changes stayed within 100 cents. Translated to quantum mechanics, this suggested that an electron's apparently random motion might reflect temporal interference rather than spatial position. Two time dimensions t_1, t_2 with one spatial dimension x described some behaviour well but lacked the fine-structure constant α, and a third time dimension, t_3, was added to carry it.

The first edition reached hydrogen's ground state with three fitted times. This edition keeps the premise and changes the machinery:

1. It identifies the two clocks the premise needs (Section 3).
2. It supplies the quantization condition the first edition described only in words, and shows that the resulting relation is exact for exactly one force law.
3. It extends the construction from hydrogen to all 118 elements (Section 5).
4. It adds the clocks that end atomic states: spontaneous emission, which runs on a cube of time, and alpha decay, which runs on the Sommerfeld ratio (Section 6).
5. It grades every prediction against literature values, with no parameter adjusted to any atom.

The companion interactive instrument, the Chronowave Temporal Engine (Module VIII), runs these calculations live, element by element. The scripts that produced every number are provided with the paper.

2 What the first edition established, and what it left open

The first edition's instincts were sound; its machinery was incomplete. Each point below states what the first edition says and the correction this edition makes.

A1. Normalization (1st-ed. eq. (5)–(6)). The wavefunction $\Psi = A\exp[i(\omega_1 t_1 + \omega_2 t_2 + \omega_3 t_3)]$ is a pure phase, so $|\Psi|^2 = A^2$ is constant, and the normalization integral over all of (t_1, t_2, t_3) diverges for every $A \neq 0$. *Fix:* normalize over one temporal cell of volume $V_T = t_1 t_2 t_3$, so that $A = V_T^{-1/2}$. A constant normalization then requires a constant cell volume.

A2. The quantization condition (1st-ed. eq. (27)). $E_n = \hbar\omega_n$, with ω_n "the allowed temporal frequencies that produce constructive interference", names a condition without writing it down. No operator, boundary condition or potential selects the ω_n. *Fix:* Section 3 supplies the Dirac bound-state condition and its temporal reading.

A3. Tables 1, 3 and 4. Tables 1 and 4 are placeholders with every entry zero. In Table 3 the energies $E_1 = h \times 1$ Hz and $E_2 = h \times 1.5$ Hz are the same in every row, so the table shows no dependence on t_1, t_2 or x. The first edition's text describes x as "angular rotation in metres", which mixes an angle with a length. Computed tables replace all three.

A4. The refinement ladder (Table 5). The Initial, Secondary, Tertiary and Final rows adjust t_1, t_2, t_3 against the known energy [FITTED]. The first three rows list the same predicted energy, 2.18×10^{-18} J, while their errors fall from 5174% to 0.0000431%. The Tertiary and Final rows share the error digit for digit although their times differ. The Final row's energy entry, 2.07×10^{16}, equals $2.18 \times 10^{-18}\,\mathrm{J}/\hbar$ in rad/s. The last step therefore changed the *representation* (energy re-read as a temporal rate), not the fit. Section 7 places these times on the derived clock ladder.

A5. The worked error example (1st-ed. eq. (29)). The arithmetic $|2.18 - 2.179|/2.179 = 0.046\%$ is correct, but it disagrees with the 0.0000431% in Table 5. The procedure text also refers to Equation (1) where it means Equation (28). This edition reports signed errors against sourced reference values.

A6. Two findings in Table 5. The Initial row sets $t_1 = 1.29 \times 10^{-21}$ s. That is the electron's Compton time, $\hbar/mc^2 = 1.288 \times 10^{-21}$ s, to the three digits given, as the first edition's text says ("the Compton time of the electron as a baseline"). The first edition began where this one does. Table 5's pair $(\omega_1, \omega_2) = (3.29, 6.58)$ has no units in the first edition. Read in units of 10^{15} Hz, which is our inference, it is the Rydberg frequency $cR_\infty = 3.2898 \times 10^{15}$ Hz and its octave.

A7. Spin sector (1st-ed. eq. (7)–(23)). This is retained in Section 8 with three corrections, to 1st-ed. eq. (9) and (23), to (11)–(13), and to the phase equation of the Lagrangian (20).

3 Methodology: two clocks, one phase

3.1 The premise, and what makes it exact

The electron carries a clock of its own. This is the phase of its rest-energy oscillation, which advances at the Compton frequency $\omega_C = mc^2/\hbar$ in its *proper* time τ. A stationary state carries a second clock, its energy phase, which advances at $E/\hbar$ in *coordinate* time t. Setting the two phases equal gives

$$\underbrace{\omega_C\,\tau}_{\text{proper clock}} \;=\; \underbrace{\frac{E}{\hbar}\,t}_{\text{coordinate clock}} \qquad\Longrightarrow\qquad E \;=\; mc^2\left\langle \frac{d\tau}{dt} \right\rangle. \tag{1}$$

Equation (1) defines what the energy means in temporal terms [FORCED-BY-DEFINITION]. Whether it holds is a question about dynamics. For the Dirac Hamiltonian $H = c\,\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2 + V(r)$, scaling

$r \to \lambda r$ gives the Dirac virial theorem, $\langle c\,\boldsymbol{\alpha}\cdot\mathbf{p}\rangle = \langle rV'\rangle$ [1], and therefore

$$E \;=\; mc^2\langle\beta\rangle + \langle V + rV'\rangle \qquad \text{[FORCED-BY-MECHANICS]}. \tag{2}$$

For a power-law potential $V \propto r^k$ the last term equals $(k+1)\langle V\rangle$. It vanishes for every bound state if and only if $k = -1$. So $E = mc^2\langle\beta\rangle$ holds exactly for the inverse-square force and for no other. For a free wave packet $\langle\beta\rangle = 1/\gamma = d\tau/dt$, and in a bound state we *identify* $\langle\beta\rangle$ with the electron's average clock rate, $\langle d\tau/dt\rangle \equiv \langle\beta\rangle$ [FORCED-BY-DEFINITION]. Proper time has no operator in quantum mechanics, so this is a reading, not a derivation. The companion calculations check the dynamical statement by integrating the radial Dirac equations: $\langle\beta\rangle$ equals E/mc^2 to 6×10^{-11} over 45 states with $Z = 1$–92, and for $V \propto r^{-0.9}, \ldots, r^{-0.25}$ the mismatch equals $(k+1)\langle V\rangle$ to every printed digit.

3.2 The quantization, and every level as a clock rate

The bound-state condition of the Dirac–Coulomb problem supplies what 1st-ed. eq. (27) left in words. Every level (n, j) of a one-electron atom has the clock rate

$$\left\langle\frac{d\tau}{dt}\right\rangle_{nj} = \left[1 + \left(\frac{Z\alpha}{n - (j+\tfrac{1}{2}) + \sqrt{(j+\tfrac{1}{2})^2 - (Z\alpha)^2}}\right)^2\right]^{-1/2} \xrightarrow{\;j=n-\frac{1}{2}\;} \sqrt{1-\left(\frac{Z\alpha}{n}\right)^2}. \tag{3}$$

For circular states ($j = n - \frac{1}{2}$) the level coincides with the relativistic Sommerfeld orbit, whose speed is exactly $\beta = Z\alpha/n$ [2]. The binding energy is the proper time the electron loses, and a spectral line is the difference of two clock rates:

$$B_{nj} = mc^2\big(1 - \langle d\tau/dt\rangle_{nj}\big), \qquad \hbar\omega_{u\to l} = mc^2\big(\langle d\tau/dt\rangle_u - \langle d\tau/dt\rangle_l\big). \tag{4}$$

Hydrogen's ground-state electron loses 2.30 s of proper time per day of coordinate time; in U^{91+} it loses 6.2 hours. For a point nucleus the 1s clock stops at $Z\alpha = 1$, that is $Z = 137.04$. A finite nucleus keeps the level bound beyond that. In our uniform-sphere model the energy passes zero near $Z \approx 147$ and $\langle\beta\rangle$ near $Z \approx 153$; the two differ because $E = mc^2\langle\beta\rangle$ fails inside a finite nucleus. The level reaches the negative-energy continuum near $Z \approx 170$ [3].

3.3 The first edition's interference, with physical frequencies

The first edition's eq. (26), $\psi = 2A\cos\big(\frac{\omega_1+\omega_2}{2}t\big)\cos\big(\frac{\omega_1-\omega_2}{2}t\big)e^{ikx}$, becomes physical when ω_1 is a free electron's clock ω_C and ω_2 is the bound state's clock $E/\hbar$, both read on the laboratory clock. By eq. (1) the bound electron's own proper clock never beats against its energy phase. What beats is the bound clock against the free one. The envelope then oscillates at B/h, so the binding energy is the beat between a free and a bound electron. For hydrogen this beat is the Lyman limit, the photon that ionizes the atom. The carrier-to-envelope ratio is $(1+f)/(1-f)$ with $f = \langle d\tau/dt\rangle$: about 75,000 for hydrogen and 6.73 for U^{91+} (Figure 1).

3.4 Many electrons

The whole atom. Consider any mean-field description in which every interaction is homogeneous of degree -1 in the coordinates. Dirac–Hartree–Fock and exchange-only Dirac–Slater theory are examples, and the frequency-independent Breit interaction is also of degree -1. In such a model,

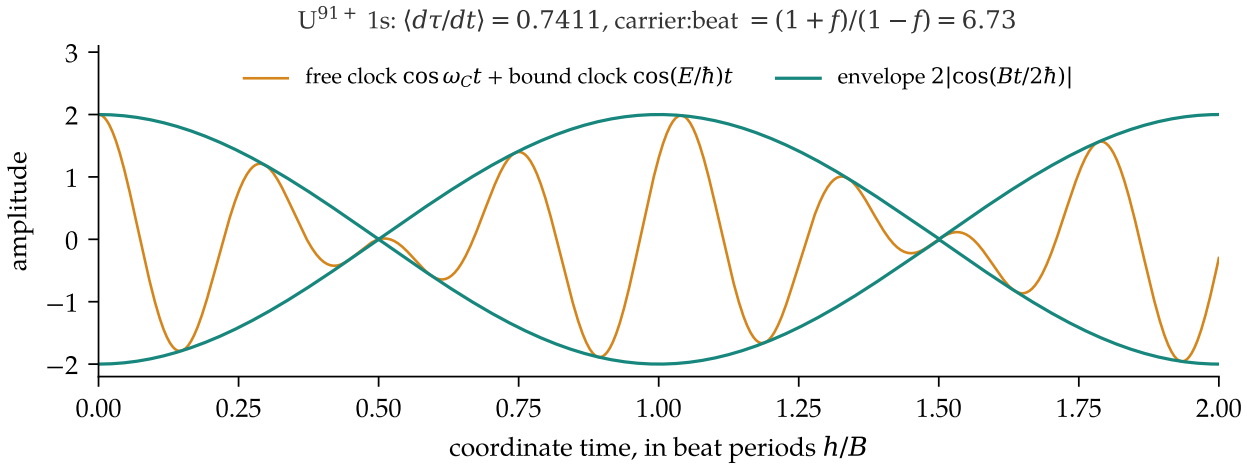


Figure 1: The first edition's Figure 1 rebuilt with physical frequencies: a free electron's clock and a bound state's clock, superposed as in its eq. (26). The envelope beats at B/h. Shown for U^{91+}, where the carrier-to-beat ratio is small enough to draw.

scaling all coordinates together gives $\langle \sum_i c\,\boldsymbol{\alpha}_i \cdot \mathbf{p}_i \rangle = -\langle V \rangle$, and the premise holds for the whole atom:

$$E_{\text{atom}} \;=\; \sum_{i=1}^{Z} mc^2 \Big(\langle d\tau/dt \rangle_i - 1 \Big) \;+\; E_{\text{inside nucleus}} \tag{5}$$

[FORCED-BY-MECHANICS, WITHIN THAT CLASS]. Inside the class, the one departure from inverse-square forces is the interior of the nucleus. There the charge is spread out and the force grows linearly, so this term is carried separately. It is 7×10^{-10} of the total energy for hydrogen, 6.5×10^{-6} for iron, 9×10^{-4} for uranium and 6×10^{-3} for oganesson. The identity does *not* hold for density functionals with correlation (which are not homogeneous), for no-pair projected many-body Hamiltonians, or for hyperfine and QED potentials.

Each electron: clock, shelter and exchange. For one electron in the atom's mean field, eq. (2) reads

$$\varepsilon_i = \underbrace{mc^2\big(\langle d\tau/dt \rangle_i - 1\big)}_{\text{clock}} + \underbrace{\big\langle U_{\text{out}} \big\rangle_i}_{\text{shelter}} + \underbrace{\big\langle \partial_r (r V_{xc}) \big\rangle_i}_{\text{xc}}. \tag{6}$$

Here $U_{\text{out}}(r) = \partial_r [r V_{\text{es}}(r)]$ is exactly the electrostatic potential energy due to all charge lying farther from the nucleus than r. That charge is the outer electrons and, for the part of an orbital inside the nucleus, the nuclear charge beyond r. By the shell theorem this charge exerts no net force at r. Semiclassically, then, it does not change the electron's local speed, and so it does not change its clock; it only lifts the energy. The clock is set by the charge inside, the shelter by the charge outside. The shelter also includes the electron's own cloud beyond r. Exact exchange would remove that self part; the local-density approximation removes only part of it. For hydrogen the model removes 2.7 of 8.1 eV, which is the familiar self-interaction error, and it is why the model's hydrogen 1s clock reads 2.15 s/day against the exact 2.30.

The computation. Each element $Z = 1$–118 was computed as a self-consistent relativistic atom:

- **Physics:** the radial Dirac equation for every electron, in a uniform-sphere nucleus (radii in Appendix A).

- **Mean field:** the Hartree field of the electrons' own density plus spin-polarized local-density exchange and correlation (Kohn–Sham [4], Perdew–Zunger [5]) [ASSUMED].
- **Inputs:** Z and the NIST ground configuration [MEASURED]; beyond $Z = 103$ the configuration is a prediction [ASSUMED]. Nothing is fitted to any atom.
- **Subshells:** each shell is split into its two j subshells by statistical weight, except that the 6p and 7p shells are filled $j = \frac{1}{2}$ first (jj coupling) [ASSUMED].
- **Ionization and K-shell binding:** Slater's transition state [6], the orbital energy at half occupation (Janak's theorem [7]).
- **The departing electron:** one electron is removed from each of the three least-bound shells in turn, and the smallest ionization energy is kept [ASSUMED]. Ion ground states that require rearranging the remaining electrons (V^+ $3d^4$, Co^+ $3d^8$, Ni^+ $3d^9$, La^+ $5d^2$, Ce^+ $4f5d^2$) lie outside this rule.

The ledger (5) is checked in the exchange-only (Dirac–Slater) version of the same atoms, where every interaction is homogeneous and the theorem holds identically.

3.5 Decay clocks

Radiative decay: a cube of time. The electric-dipole emission rate $A = 4\alpha\omega^3|\mathbf{r}_{fi}|^2/3c^2$, rewritten in time and nothing else, is

$$\tau_{E1} = \frac{1}{A} = \frac{3}{32\pi^3\alpha}\,\frac{T^3}{t_d^2}, \qquad t_d = \frac{|\mathbf{r}_{fi}|}{c} \tag{7}$$

[FORCED-BY-MECHANICS]. Here T is the period of the emitted light and t_d is the light-crossing time of the transition dipole, with $|\mathbf{r}_{fi}|^2$ summed over final sublevels. For a p→s line, $|\mathbf{r}_{fi}|^2 = R^2/3$, where R is the radial integral. The lifetime is thus the cube of the period divided by the square of the dipole time, scaled by $1/\alpha$. For one-electron ions $T \propto Z^{-2}$ and $t_d \propto Z^{-1}$, so $\tau \propto Z^{-6}/Z^{-2} = Z^{-4}$, and the excited states of heavy one-electron ions die fastest.

Alpha decay: the Sommerfeld count. An alpha particle (charge $Z_2 = 2$) with asymptotic speed βc escapes a daughter nucleus of charge Z_1 and mass number A_d through the Coulomb barrier. It does so after e^{2G} ticks of a nuclear clock $t_a = 2R/\beta c$ [8, 9]:

$$t_{1/2} = \ln 2\; t_a\, e^{2G}, \qquad 2G = 4\eta\left[\arccos\sqrt{x} - \sqrt{x(1-x)}\right] \xrightarrow{R\to 0} 2\pi\eta, \qquad \eta = \frac{Z_1 Z_2\,\alpha}{\beta}. \tag{8}$$

Here $x = R/b$, and $b = Z_1 Z_2 e^2/(4\pi\varepsilon_0 Q)$ is the classical turning radius. The Sommerfeld number η is the Coulomb speed divided by the particle's speed; for a bound electron on a circular Sommerfeld orbit it equals n. We take the textbook radius $R = 1.2\,(A_d^{1/3} + 4^{1/3})$ fm and the assault time at the asymptotic speed [ASSUMED].

3.6 Error metric

The first edition's eq. (28) is kept, but errors are reported signed. Grading targets are literature values, and none is used as an input. Some are measurements: the deuterium line, the X-ray lines,

the K-shell binding energies, the alkali lifetimes and the half-lives. Hydrogen's 2p lifetime is a theoretical reference value. Others are NIST reference values: the ionization energies, which for H and He^+ come from QED theory anchored to spectroscopy.

4 Results: one-electron atoms

4.1 Hydrogen, deuterium, He^+ and U^{91+}

Table 1 and Figure 2 compare the non-relativistic (Bohr) atom with the proper-time atom of eq. (3), using reduced masses and CODATA constants. In every case the proper-time atom leaves only the Lamb shift: radiative QED, which is beyond any clock-rate construction, plus the finite nuclear size for heavy ions. The Lamb shifts used are the measured 1S shifts of hydrogen (8172.9 MHz) and deuterium (8183.8 MHz) [10]. For He^+ the hydrogen value is scaled by the leading-order QED law $Z^4[\ln(Z\alpha)^{-2} - \ln k_0 + \frac{19}{30}]$, with the 1S Bethe logarithm $\ln k_0 = 2.984$.

Table 1: One-electron atoms with zero free parameters. Errors are prediction minus reference.

target (reference)	Bohr	proper time	Lamb shift	after it
H ionization, 13.5984346 eV (NIST)	−10.83 ppm	+2.48 ppm	2.49 ppm	−0.01 ppm
He^+ ionization, 54.4177655 eV (NIST)	−45.08 ppm	+8.18 ppm	8.10 ppm	+0.08 ppm
D Lyα 1S–$2P_{3/2}$, 121.53375 nm [11]	+13.38 ppm	−3.26 ppm	−3.32 ppm	+0.05 ppm
U^{91+} 1s binding [13]	−12.64%	+0.35%	0.35%	by definition

Herzberg's deuterium line. Herzberg [11] measured the unresolved deuterium Lyman-α doublet at 1215.3378 ± 0.0003 Å and reported a ground-state Lamb shift of 0.262 ± 0.038 cm^{-1}. A −0.0003 Å profile correction gives 1215.3375 Å for the 1S–$2P_{3/2}$ component, as reconstructed in [12]. Equation (3), with the reduced mass, gives 82 281.939 cm^{-1} for this transition, the Dirac value of that analysis. The 0.268 cm^{-1} gap to Herzberg's line is his ground-state Lamb shift. Adding the measured 1S Lamb shift [10] closes it to +0.05 ppm, within his ±0.0003 Å (±0.25 ppm).

U^{91+}. U^{91+} sets a scale rather than a test: its measured quantity is the Lamb shift itself, 460.2 ± 4.6 eV [13], defined against this very Dirac value. Bohr misses 12.6% of the binding of the heaviest one-electron atom measured. The proper-time atom (point nucleus) misses 0.35%, which is the whole measured Lamb shift. Of that, about 0.15% is the finite size of the nucleus and about 0.2% is QED.

The earlier residuals. The non-relativistic residuals are now fully explained. Their Dirac part is −13.31 ppm in H and −53.25 ppm in He^+, a ratio of $4.000 = Z^2$. The Lamb shift grows as $Z^4 \ln(Z\alpha)^{-2}$, a little more slowly than the Dirac term's Z^4, so it cancels 19% of the Dirac term in H but only 15% in He^+. Dirac plus Lamb therefore predicts the He^+/H residual ratio as 4.17; the reference values give 4.16.

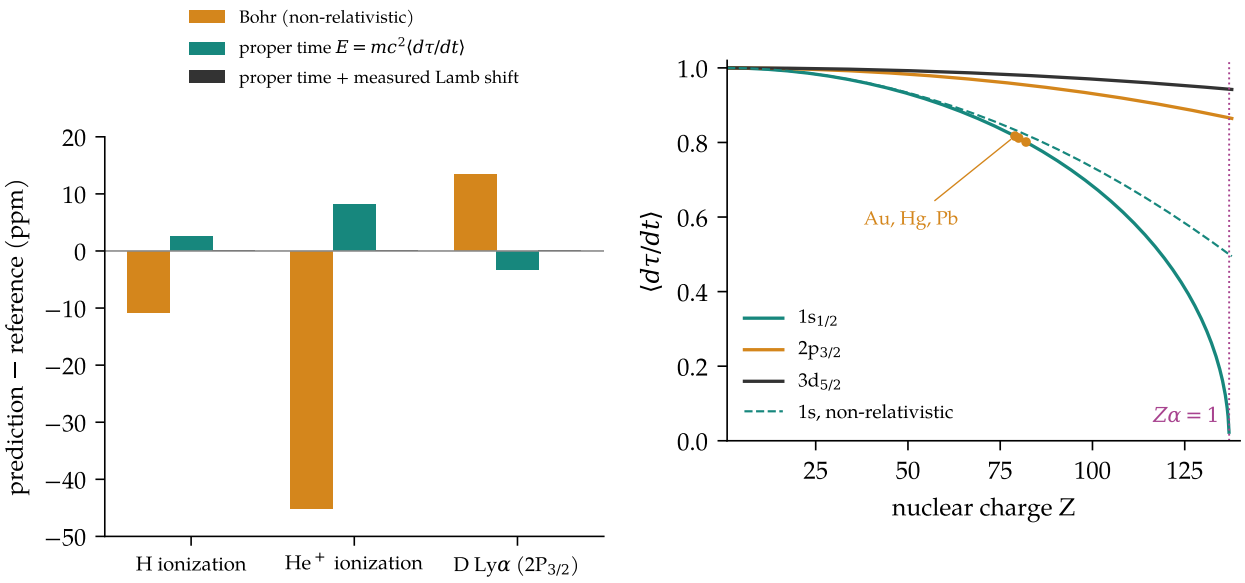


Figure 2: Left: one-electron predictions against reference values; the proper-time atom leaves only the Lamb shift. Right: $\langle d\tau/dt\rangle$ across the table for a point nucleus. The Dirac 1s clock stops at $Z\alpha = 1$. The non-relativistic curve does not stop within the table (it would reach zero only at $Z \approx 194$). Markers: gold, mercury and lead, whose 1s clocks run at 80%–82% speed.

4.2 X-ray lines of real atoms

Moseley's law [14], read as a difference of clock rates, is $K\alpha_1 = mc^2[\langle d\tau/dt\rangle_{2p_{3/2}} - \langle d\tau/dt\rangle_{1s}]$ at $Z - 1$, with Moseley's screening $\sigma = 1$ [ASSUMED]. Against 32 $K\alpha_1$ lines from Al to U [15], the RMS error falls from 5.71% (Bohr/Moseley) to 1.05% (Figure 3a). Moseley's straight line bends away from the data as Z grows, by 14% at uranium, and that bend is the proper-time deficit the non-relativistic law leaves out. The remainder, -0.9% to $+1.5\%$ and mostly $+0.4\%$ to $+1.5\%$, reflects outer-electron screening. The $K\alpha$ doublet is a purely relativistic effect: the non-relativistic atom predicts a splitting of exactly zero at every Z. With Sommerfeld's doublet screening $\sigma = 3.5$ [ASSUMED], it comes out within 0.66% for $Z \geq 40$, over splittings from 84 to 3774 eV (Figure 3b). Below $Z \approx 30$ a single screening constant fails (-14% at Al).

5 Results: all 118 elements

5.1 The whole-atom ledger

For every element from hydrogen to oganesson, the exchange-only Dirac–Slater atom satisfies eq. (5) to 1.1×10^{-7} or better, with a nuclear-volume term of at most 0.6% (oganesson). The total energy of each of these atoms is the sum of its electrons' proper-time deficits plus that term. The closure checks an identity of the model; it is not a measure of the model's accuracy. With correlation included, the identity picks up a residual from the correlation functional, which is not homogeneous: 2.5% of the total for hydrogen, 0.5% for nitrogen, and about 3×10^{-4} by gold.

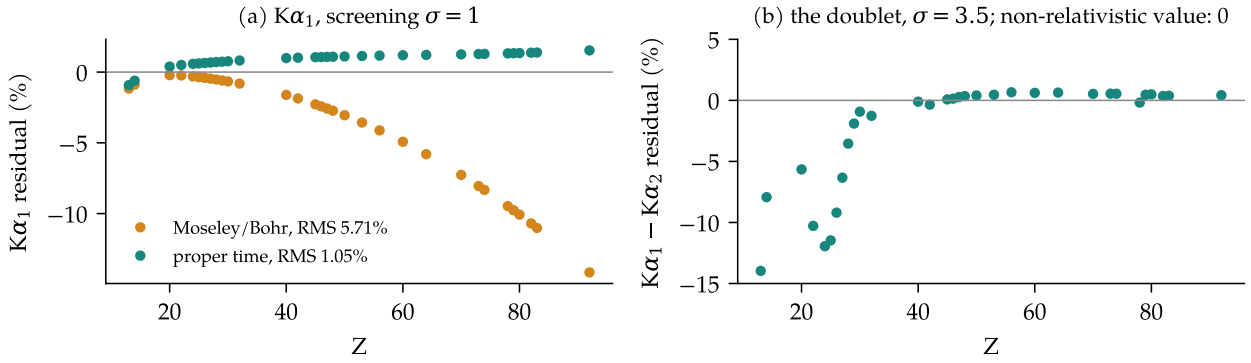


Figure 3: (a) Kα_1 residuals, Bohr/Moseley against proper time. (b) The Kα doublet, which is zero in any non-relativistic atom. Line energies from the X-ray Data Booklet, Table 1-2.

5.2 Ionization and K-shell binding

Figure 4 and Appendix C compare the computed first ionization energies with the reference values. Over the 99 elements with a NIST reference value that is not merely an estimate, the RMS error is 5.0% and the median absolute error is 3.2%. For hydrogen the NIST value is QED theory anchored to spectroscopy. The rule for the departing electron gives the known departing shell where one exists. Lutetium, hafnium, yttrium and palladium lose a d electron (Lu^+ is $4f^{14}6s^2$), and actinium, protactinium, uranium, neptunium and curium lose a 6d. The N/O inversion of the half-filled shell is reproduced; the P/S and As/Se inversions are not.

The largest misses, 8–20%, are transition metals: Ni, Cr, Cu, Co, Pd, Nb, Pt, Ag, Ru, Mo and Rh. Local-density exchange is known to misplace the balance between d and s electrons. For nickel most of the +19.5% has a second cause: its true ion ground state ($3d^9$) requires rearranging the remaining electrons, which the one-electron removal cannot reach. For $Z = 104$–118 no measurement exists, and the comparison is with theoretical estimates (marked † in Appendix C).

K-shell binding across the 83 elements from neon to uranium [15] agrees to 0.46% RMS, with a mean of +0.42% (Figure 5, left). Above $Z \approx 40$ the residual rises smoothly to +0.63% at uranium. That is the size and sign expected from the Breit interaction and QED, which reduce K-shell binding by a few hundred eV at uranium and are absent from a Dirac mean field. Below $Z \approx 40$, where those effects are $\lesssim$0.1%, the residual reaches +1.2% (sodium). There it mainly reflects the comparison itself: the Data Booklet references metals to the Fermi level rather than to a free atom's vacuum level, and the transition-state treatment adds its own error. For H and He the K-shell binding is the first ionization energy.

5.3 What the ledger shows

The ledger of eq. (6) has the same shape across the table. The electron that ionizes loses between about one second (lithium's 2s) and 36 seconds (copernicium's 6d) of proper time per day, half to fifteen times the exact hydrogen value of 2.30 s. It dips through the core, where the enclosed charge is large. What makes it loosely held is shelter: the potential of the charge lying outside it cancels about two-thirds to three-quarters of its clock term for the alkali s electrons (66% for Li to 76% for Fr), and up to 97% for palladium's 4d. The exchange–correlation term is not small either; for the ionizing electron its median size is half of $|\varepsilon|$. The binding of the outer electron is therefore a small residue of a large clock term and a large shelter term, with a smaller exchange–correlation term,

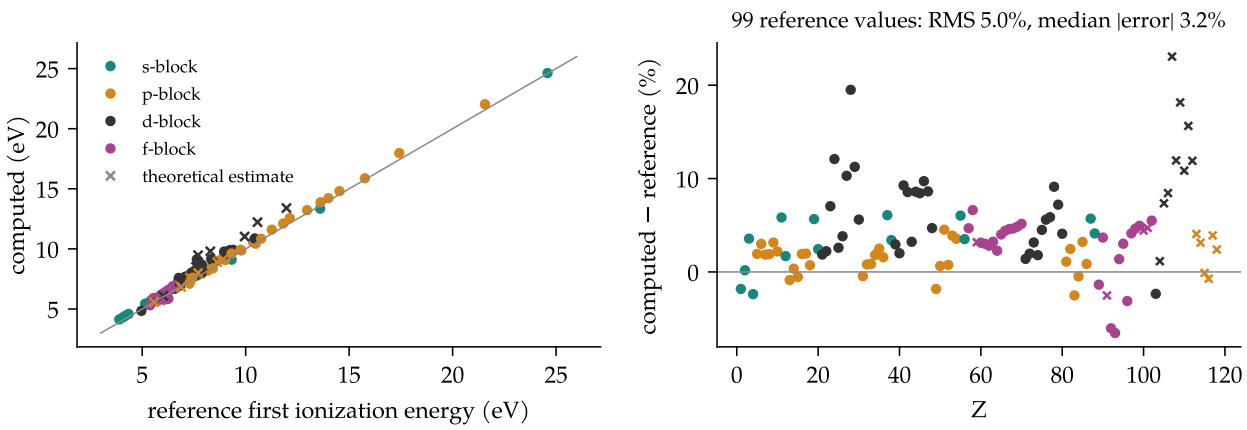


Figure 4: First ionization energies for all 118 elements, computed with no per-atom parameters, against NIST reference values ($Z \leq 108$) and theoretical estimates (crosses; $Z \geq 109$ and the NIST estimates).

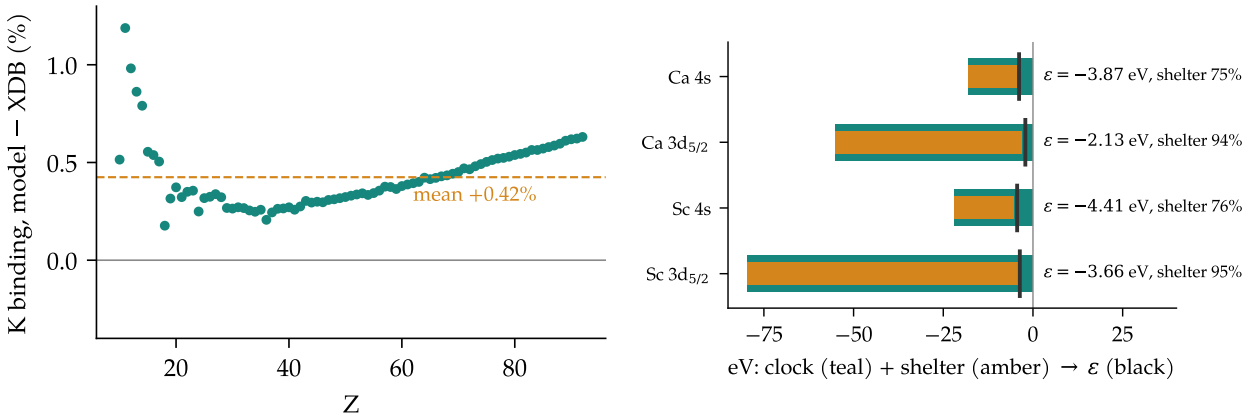


Figure 5: Left: K-shell binding energies against the X-ray Data Booklet. Right: the clock-plus-shelter ledger of the 4s and $3d_{5/2}$ orbitals of calcium and scandium, in the spin-up channel of each neutral atom's self-consistent potential. Calcium's 3d is evaluated unoccupied in calcium's own field.

which is why chemistry is so sensitive.

The ledger also shows what orders calcium's 4s and 3d orbitals. In calcium's field the 3d is the *slower* clock of the two. Its clock term, −55.2 eV, is 3.05 times the 4s's −18.1 eV, because it sits deeper, at $\langle r \rangle = 2.3$ against 4.0 bohr. But 94% of that clock is sheltered, against 75% for the 4s, so the 4s orbital lies 1.7 eV lower (Figure 5, right). The ordering of the two orbital energies is set by shelter, not by the clock. Which orbitals are occupied is decided by total energies, not orbital energies alone. In scandium the occupied 3d still lies 0.75 eV above the 4s (spin-up channel), the familiar 4s/3d paradox, which the ledger describes but does not resolve.

At the heavy end, the 1s clock of oganesson loses 11.3 hours per day. From rutherfordium to copernicium the 7s is pulled inward where clocks are slow. In this model the 6d is the electron that ionizes in every one of them except bohrium, where 7s wins by 0.1 eV. This is a prediction; none of these ions has been measured.

6 Results: the decay clocks

6.1 Spontaneous emission and the cube of the period

For hydrogen's 2p state, eq. (7) with the measured period $T = 0.4055$ fs and $t_d = 0.1315$ as gives 1.598 ns (1.595 ns with the Dirac period), against 1.596 ns from the NIST A-value 6.2649×10^8 s^{-1}. For the alkali resonance lines (Table 2), the dipole comes from the computed atoms and the period from the measured line. Each of the ground and excited states is solved self-consistently. The predicted lifetimes land within +0.6% (Li), +1.1% (Na), −3.9% (K), −8.0% (Rb) and −8.1% (Cs) of the NIST Handbook values [16]. The Handbook's Rb value is older; more recent measurements are closer to 26.2 ns, which would move Rb to about −5% [19]. The heavy-alkali shortfall is core polarization, which a mean field omits. Because the period enters cubed, a 3% error in the line energy becomes a 9% error in the lifetime. That is why the fully computed version, which uses the model's own period, drifts by up to −18%. Figure 6 shows the Z^{-4} law for one-electron ions, from nanoseconds in hydrogen to 7 attoseconds in the heaviest ions. Neutral atoms' valence lifetimes do not follow it; the alkalis all sit at 16–31 ns.

Table 2: The cube law against reference lifetimes. R: radial dipole integral (bohr); T: measured period (fs); $t_d = R/(\sqrt{3}\,c)$ (as); lifetimes in ns; error uses the measured T.

line	R	T	t_d	τ (meas. T)	τ (model T)	τ (NIST)	error %
H $2p_{3/2}$	1.2903	0.4055	0.1315	1.598	1.595	1.596	+0.1
Li $2p_{3/2}$	4.050	2.238	0.413	27.27	28.69	27.10	+0.6
Na $3p_{3/2}$	4.295	1.965	0.438	16.42	14.57	16.23	+1.1
K $4p_{3/2}$	5.130	2.557	0.523	25.35	23.03	26.39	−3.9
Rb $5p_{3/2}$	5.319	2.603	0.542	24.86	22.05	27.03	−8.0
Cs $6p_{3/2}$	5.715	2.843	0.582	28.07	25.64	30.53	−8.1

Where the first edition's t_3 may live. [OPEN] The fine-structure constant α enters eq. (7) exactly once, as the coupling between the period's cube and the dipole's square. That is the role the first edition gave t_3. Its "Scramble Zone", picoseconds to nanoseconds, is also where optical spontaneous emission sits: 1.6 ns for hydrogen, 16–31 ns for the alkalis. Whether the paper's third time dimension is this T^3 remains open. We claim only that the cube is real and say where it appears.

6.2 Alpha decay

Across 35 even-even emitters from Te-106 to Og-294 (Appendix D), eq. (8) with $r_0 = 1.2$ fm and no fitted constant predicts half-lives spanning 29.8 decades. The RMS error is 0.56 decades and the mean offset is −0.39 decades. The absolute level depends on the assumed radius: $r_0 = 1.15$, 1.25 and 1.3 fm give RMS errors of 0.45, 1.11 and 1.70 decades. The scatter about a single overall offset does not: it stays at 0.37–0.46 decades for r_0 between 1.15 and 1.3 fm. One overall constant, standing for the alpha-formation probability, gives 0.40 decades at $r_0 = 1.2$ fm [FITTED], but the result does not need it. Po-210 is the one outlier, at −1.6 decades raw and −1.2 after the offset. It sits on the $N = 126$ shell closure, where alpha particles are known to form less readily. The nuclear clock $t_a \approx 10^{-21}$ s is of the same order as the electron's Compton period, and each half-life is that tick multiplied by e^{2G}, between 10^{14} and 10^{44} ticks.

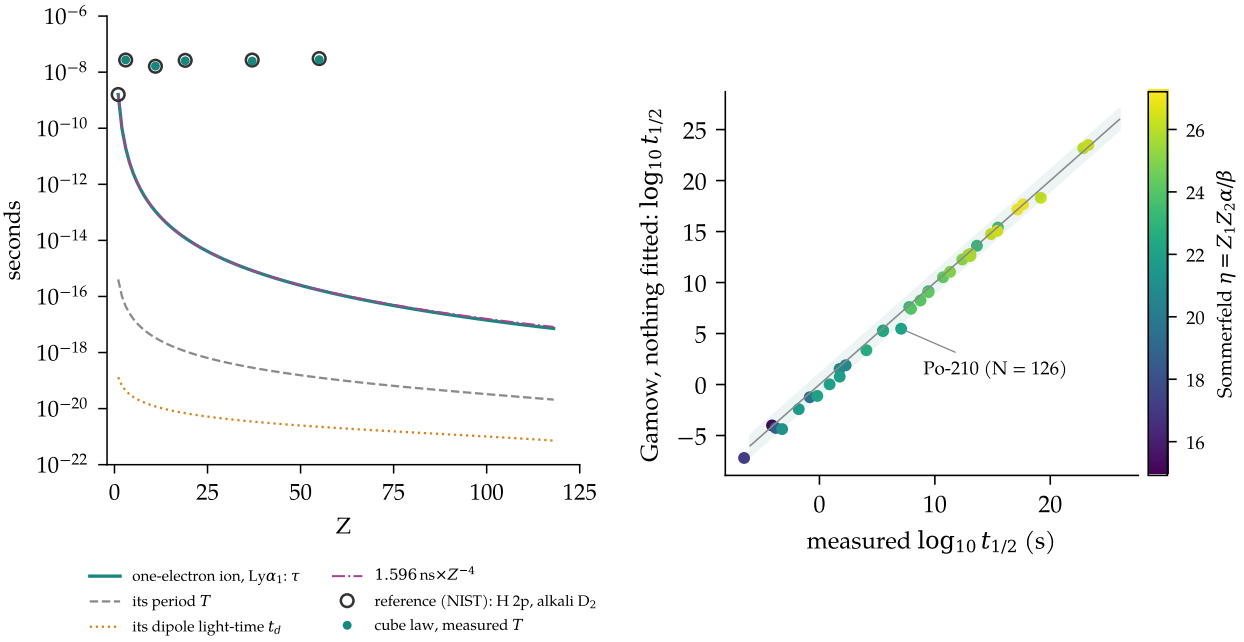


Figure 6: Left: the cube law. One-electron-ion lifetimes, with their periods and dipole times, the Z^{-4} guide, and reference lifetimes (open circles) with cube-law predictions (dots). Right: alpha-decay half-lives from eq. (8) with $r_0 = 1.2$ fm and no fitted constant, against measurement; the band is ± 1 decade.

6.3 The radioactive edge of the table

Appendix E lists the longest-lived isotope of every element with no stable isotope [MEASURED] [17]. Half-lives run from 2×10^{19} years (Bi-209) down to under a millisecond (Og-294). For the alpha emitters among them, the ratio $Z\alpha/\beta$ that sets an atom's clock also sets the decay: the decay energy Q grows beyond uranium, β grows with it, the Sommerfeld number falls, and $e^{2\pi\eta}$ collapses. Tc, Pm, At, Fr and Ac decay instead by electron capture or β decay, and Lr, Rf and Ds by fission.

7 The first edition's fitted times on the clock ladder

For hydrogen the derived clocks form a ladder whose rungs are set by successive powers of $Z\alpha$ (Figure 7):

- the Compton clock, $\hbar/mc^2 = 1.288 \times 10^{-21}$ s (α^0);
- the binding beat, $\hbar/B = 4.84 \times 10^{-17}$ s, and the orbital period, 1.52×10^{-16} s (both α^{-2}; the orbit is a full period, the others are $\hbar/E$ times);
- the fine-structure time, 1.45×10^{-11} s (α^{-4});
- the 2S–2P Lamb-shift time, 1.5×10^{-10} s.

The first edition's Initial $t_1 = 1.29 \times 10^{-21}$ s is the Compton clock. Its final t_1 and t_2 sit at round ratios to it, $t_1/t_C = 1/7.00$ and $t_2/t_C = 10.01$. This may be the fit rediscovering the proper clock through a rescaling, or a coincidence at three-digit precision [OPEN]. The final $t_3 = 5.35 \times 10^{-13}$ s matches no rung; it falls between the binding beat and the fine-structure time.

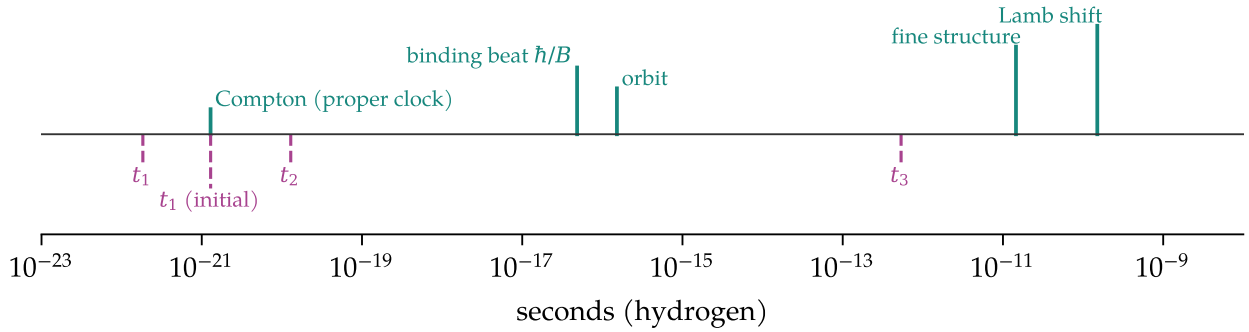


Figure 7: Hydrogen's derived clock ladder (teal) and the first edition's fitted times (magenta, dashed).

8 Spin in the cubic-time framework (retained, with corrections)

The first edition's spin sector (1st-ed. eq. (7)–(23)) is the part of the framework with dynamics of its own, and it is retained here in summary. The spin state evolves as $\chi(t_1, t_2, t_3) = \chi_0 \exp[i\omega_s(t_1 + t_2 + t_3)]$, where ω_s is the spin frequency of 1st-ed. eq. (7). The spinor of 1st-ed. eq. (19) carries the orientation θ and phase ϕ, and the Lagrangian

$$L = \tfrac{1}{2} m(t_1, t_2, t_3)\, \dot{x}^2 + \tfrac{1}{2} I(\theta)\, \dot{\theta}^2 + \lambda \cos\phi\, \dot{\theta} \qquad \text{(1st ed. 20)}$$

couples mass-energy to spin. Three corrections apply.

S1. 1st-ed. eq. (9) and (23). $\omega_p = \frac{\omega_s}{2}(t_1 - t_2)$ is dimensionally a phase, not a frequency. The accumulated angle is $\Delta\phi_p = \frac{\omega_s}{2}(t_1 - t_2)$, and the precession frequency is $\omega_p = \frac{\omega_s}{2}\frac{d}{dt}(t_1 - t_2)$. Field-free precession then requires the two time coordinates to advance at different rates. A clock-rate difference already produces a precession in standard physics, Thomas precession, so any prediction must be distinguished from it. It also needs a preferred axis, which ω_s alone does not supply [OPEN].

S2. 1st-ed. eq. (11)–(13). $\hat{S}_{t_i} = \hbar\omega_s\, \partial/\partial t_i$ has units of energy per time. An angular-momentum operator needs $\hat{S}_{t_i} = -i\hbar\, \partial/\partial(\omega_s t_i)$, whose eigenvalue on $e^{i\omega_s t_i}$ is $\hbar$.

S3. 1st-ed. eq. (22). L contains no $\dot{\phi}$, so the Euler–Lagrange equation for ϕ reduces to the constraint $\partial L/\partial\phi = -\lambda \sin\phi\, \dot{\theta} = 0$. As written, this freezes ϕ at 0 or π, or stops θ. A kinetic term $\frac{1}{2} J\dot{\phi}^2$ is needed for the phase to have dynamics [OPEN].

The "human body" application of the temporal-volume conversion, 1st-ed. eq. (24) ($V_{\text{temporal}} = V_{\text{spatial}}/t_1 t_2$, with units of m^3/s^2), lies outside the scope of this atomic paper and is not carried forward.

9 Discussion

What is new and what is not. The Dirac virial theorem, the Dirac–Coulomb spectrum, mean-field atoms and the Gamow factor are standard physics; every number here could be computed without the temporal reading. The framework contributes a labelling, and the labelling earns its place in four ways.

- It states exactly where the premise holds and where it fails.

- It turns screening into a ledger of clock, shelter and exchange terms.
- It exposes a literal cube of time in the radiative decay law.
- It connects the atom's clock ($\beta = Z\alpha/n$) to the alpha-decay clock ($\eta = Z_1 Z_2 \alpha/\beta$) through a single ratio.

What lies beyond the construction. Each of the following remains visible as a residual:

- the Lamb shift, from QED and the finite nuclear size;
- the Breit interaction;
- correlation beyond the local-density model, the largest valence errors;
- core polarization, in the alkali lifetimes.

The spin sector's field-free precession is the framework's one proposed dynamical prediction that no spatial result supplies. Stating it with correct dimensions (S1) and separating it from Thomas precession is the next step.

The constant α and t_3. The first edition proposed that t_3 sets the fine-structure coupling. The sensitivity of a line to α is $K = d\ln\omega/d\ln\alpha$. In the non-relativistic atom every line has $K = 2$, which cannot be distinguished from a redshift or a change of units, so only the relativistic, proper-time part $K - 2$ is observable. For hydrogen's Lyman-α that part is 3.3×10^{-5}; for U^{91+} it is 0.43. Atomic clocks measure $\dot{\alpha}/\alpha = (1.0 \pm 1.1) \times 10^{-18}\ \mathrm{yr}^{-1}$ [20]. Suppose one of the time dimensions tracked cosmic age and α depended on it, with $S = d\ln\alpha/d\ln t_i$. Then α would drift at S/t_0, with $t_0 = 13.8$ Gyr, and the clock bound requires $|S| < 4.4 \times 10^{-8}$ at 2σ. The same experiment bounds α's response to gravitational potential, $(c^2/\alpha)\, d\alpha/d\Phi = (14 \pm 11) \times 10^{-9}$ [20].

10 Conclusion

The first edition asked whether an electron's states could be found with time. With two clocks and no per-atom fitting, they can: exactly for one-electron atoms, and for every element within a mean-field model. Locking the electron's clock to the state's clock gives $E = mc^2\langle d\tau/dt\rangle$, exact for the inverse-square force. It reproduces hydrogen, deuterium and He^+ to the Lamb shift and the X-ray lines to 1%. Summed over electrons, it gives the total energy of every mean-field atom in the periodic table, as an identity closed to 10^{-7} once the nuclear-volume term is included. With shelter and exchange added, it gives ionization energies to 5% and K-shell binding energies to 0.5% for $Z \geq 10$. The same temporal language describes how atomic states end: by a cube of the light's period, and by the Sommerfeld count at the nuclear barrier. The first edition's intuition that time can carry the atom was right; what it needed was the second clock.

A Numerical methods and checks

- **Integration.** The radial Dirac equations for $P = rg$ and $Q = rf$ are solved on a logarithmic grid of 3200 points from 10^{-7} to 60 bohr, by fourth-order Runge–Kutta shooting. Midpoint potentials use four-point Lagrange interpolation of rV. Energies come from node-count

bisection with the matching correction $\delta E = cP(r_m)[Q_{\text{out}} - Q_{\text{in}}]/\int(P^2 + Q^2)$, converged to 10^{-11}.

- **Solver validation.** Against the exact Dirac–Coulomb levels, the error is below 1.3×10^{-10} for $Z \leq 26$ and below 5×10^{-8} at $Z = 92$. The finite-nucleus 1s shift in U is 193.8 eV.
- **Self-consistency.** The potential is linearly mixed until its maximum relative change falls below 10^{-9}. All 118 atoms and all transition states converged.
- **Identity checks.** The whole-atom ledger closes to 1.1×10^{-7} in the exchange-only atoms. The per-orbital identity (6) closes to 0.05 eV absolute in orbitals bound by up to 220 keV; the residual comes from the numerical derivative of rV.
- **Nuclear model.** A uniform sphere with $R = \sqrt{5/3}\, r_{\text{rms}}$. Here $r_{\text{rms}} = 0.8414$ fm for $A = 1$, $1.0 + 0.3A^{1/3}$ fm for $A < 10$, and $0.836A^{1/3} + 0.570$ fm otherwise. A is the mass number of the standard atomic weight, or of the longest-lived isotope.
- **Alkali lifetimes.** The ground and excited configurations are each solved self-consistently (spin-polarized), and the radial dipole is taken between them.
- **Alpha decay.** Q values are taken from AME2020 atomic masses [18], cross-checked against a second tabulation. Half-lives are alpha partial half-lives where the branching is below 99.9%.

B Data sources

- **First ionization energies:** NIST Atomic Spectra Database [21] for $Z \leq 108$, and the element pages of Wikipedia for the theoretical estimates at $Z = 109$–118.
- **K-shell binding and Kα lines:** X-ray Data Booklet, Tables 1-1 and 1-2 [15].
- **Transition probabilities:** NIST Handbook of Basic Atomic Spectroscopic Data [16]. Hydrogen 2p: NIST ASD.
- **Half-lives:** NUBASE2020 [17] as quoted on the Wikipedia isotope pages, which also give the newer values used for Sm-146 (2024) and Gd-148.
- **1S Lamb shifts:** Weitz et al. [10].
- **U^{91+}:** Gumberidze et al. [13].
- **Deuterium Lyman-α:** Herzberg [11], with the component analysis of [12].

C The periodic table in proper time

Columns.

- Configuration: NIST ground state; * marks a prediction.
- IE: first ionization energy in eV, computed and reference. Marks: † theoretical estimate; ‡ NIST value from QED theory anchored to spectroscopy.
- K: K-shell binding, computed and X-ray Data Booklet, in eV (k = keV); errors for $Z \geq 10$. For H and He the K-shell binding is the first ionization energy.
- Clock: proper time lost per day by the 1s electron and by the electron that ionizes. For hydrogen the model's value, 2.15 s, carries the local-density self-interaction error; the exact value is 2.30 s.

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
1	H	$1s^1$	13.351	13.5984‡	−1.8	13.4	13.6	–	2.15 s	2.15 s
2	He	$1s^2$	24.632	24.5874	+0.2	24.6	24.6	–	6.36 s	6.36 s
3	Li	[He] $2s^1$	5.584	5.3917	+3.6	66.9	54.7	–	16.11 s	1.14 s
4	Be	[He] $2s^2$	9.101	9.3227	−2.4	126.5	111.5	–	30.31 s	2.61 s
5	B	[He] $2s^2\,2p^1$	8.458	8.2980	+1.9	204.4	188.0	–	49.07 s	3.52 s
6	C	[He] $2s^2\,2p^2$	11.599	11.2603	+3.0	300.6	284.2	–	72.34 s	5.90 s
7	N	[He] $2s^2\,2p^3$	14.804	14.5341	+1.9	415.2	409.9	–	100.13 s	8.81 s
8	O	[He] $2s^2\,2p^4$	13.878	13.6181	+1.9	550.0	543.1	–	2.2 min	11.85 s
9	F	[He] $2s^2\,2p^5$	17.972	17.4228	+3.2	703.1	696.7	–	2.8 min	15.54 s
10	Ne	[He] $2s^2\,2p^6$	22.036	21.5645	+2.2	874.7	870.2	+0.51	3.5 min	19.81 s
11	Na	[Ne] $3s^1$	5.439	5.1391	+5.8	1.084 k	1.071 k	+1.19	4.3 min	1.72 s
12	Mg	[Ne] $3s^2$	7.776	7.6462	+1.7	1.316 k	1.303 k	+0.98	5.1 min	3.15 s
13	Al	[Ne] $3s^2\,3p^1$	5.934	5.9858	−0.9	1.573 k	1.560 k	+0.86	6.0 min	3.05 s
14	Si	[Ne] $3s^2\,3p^2$	8.180	8.1517	+0.4	1.854 k	1.839 k	+0.79	7.0 min	4.84 s
15	P	[Ne] $3s^2\,3p^3$	10.430	10.4867	−0.5	2.157 k	2.146 k	+0.55	8.1 min	6.88 s
16	S	[Ne] $3s^2\,3p^4$	10.559	10.3600	+1.9	2.485 k	2.472 k	+0.54	9.3 min	8.98 s
17	Cl	[Ne] $3s^2\,3p^5$	13.222	12.9676	+2.0	2.837 k	2.822 k	+0.50	10.5 min	11.39 s
18	Ar	[Ne] $3s^2\,3p^6$	15.873	15.7596	+0.7	3.212 k	3.206 k	+0.18	11.8 min	14.08 s
19	K	[Ar] $4s^1$	4.586	4.3407	+5.7	3.620 k	3.608 k	+0.32	13.2 min	1.86 s
20	Ca	[Ar] $4s^2$	6.263	6.1132	+2.4	4.054 k	4.038 k	+0.37	14.6 min	3.05 s
21	Sc	[Ar] $3d^1\,4s^2$	6.684	6.5615	+1.9	4.506 k	4.492 k	+0.32	16.2 min	3.57 s
22	Ti	[Ar] $3d^2\,4s^2$	6.979	6.8281	+2.2	4.983 k	4.966 k	+0.35	17.8 min	3.97 s
23	V	[Ar] $3d^3\,4s^2$	7.221	6.7462	+7.0	5.484 k	5.465 k	+0.36	19.5 min	4.31 s
24	Cr	[Ar] $3d^5\,4s^1$	7.585	6.7665	+12.1	6.004 k	5.989 k	+0.25	21.3 min	4.31 s
25	Mn	[Ar] $3d^5\,4s^2$	7.627	7.4340	+2.6	6.560 k	6.539 k	+0.32	23.2 min	4.91 s
26	Fe	[Ar] $3d^6\,4s^2$	8.204	7.9025	+3.8	7.135 k	7.112 k	+0.32	25.1 min	5.33 s
27	Co	[Ar] $3d^7\,4s^2$	8.692	7.8810	+10.3	7.735 k	7.709 k	+0.34	27.1 min	5.71 s
28	Ni	[Ar] $3d^8\,4s^2$	9.130	7.6399	+19.5	8.360 k	8.333 k	+0.32	29.2 min	6.05 s
29	Cu	[Ar] $3d^{10}\,4s^1$	8.596	7.7264	+11.3	9.003 k	8.979 k	+0.27	31.4 min	5.37 s
30	Zn	[Ar] $3d^{10}\,4s^2$	9.921	9.3942	+5.6	9.685 k	9.659 k	+0.26	33.7 min	6.70 s
31	Ga	[Ar] $3d^{10}\,4s^2\,4p^1$	5.972	5.9993	−0.5	10.395 k	10.367 k	+0.27	36.0 min	4.27 s
32	Ge	[Ar] $3d^{10}\,4s^2\,4p^2$	7.963	7.8994	+0.8	11.133 k	11.103 k	+0.27	38.5 min	6.21 s
33	As	[Ar] $3d^{10}\,4s^2\,4p^3$	9.872	9.7886	+0.9	11.897 k	11.867 k	+0.25	41.0 min	8.29 s
34	Se	[Ar] $3d^{10}\,4s^2\,4p^4$	9.930	9.7524	+1.8	12.689 k	12.658 k	+0.25	43.6 min	10.37 s
35	Br	[Ar] $3d^{10}\,4s^2\,4p^5$	12.109	11.8138	+2.5	13.509 k	13.474 k	+0.26	46.3 min	12.64 s
36	Kr	[Ar] $3d^{10}\,4s^2\,4p^6$	14.220	13.9996	+1.6	14.356 k	14.326 k	+0.21	49.0 min	15.07 s
37	Rb	[Kr] $5s^1$	4.431	4.1771	+6.1	15.237 k	15.200 k	+0.24	51.9 min	2.29 s
38	Sr	[Kr] $5s^2$	5.889	5.6949	+3.4	16.147 k	16.105 k	+0.26	54.8 min	3.54 s
39	Y	[Kr] $4d^1\,5s^2$	6.400	6.2173	+2.9	17.083 k	17.038 k	+0.26	57.8 min	9.96 s
40	Zr	[Kr] $4d^2\,5s^2$	6.767	6.6341	+2.0	18.047 k	17.998 k	+0.27	61.0 min	4.89 s
41	Nb	[Kr] $4d^4\,5s^1$	7.385	6.7588	+9.3	19.035 k	18.986 k	+0.26	64.2 min	5.18 s
42	Mo	[Kr] $4d^5\,5s^1$	7.699	7.0924	+8.6	20.055 k	20.000 k	+0.27	67.4 min	5.62 s
43	Tc	[Kr] $4d^5\,5s^2$	7.349	7.1194	+3.2	21.108 k	21.044 k	+0.30	70.8 min	6.11 s
44	Ru	[Kr] $4d^7\,5s^1$	7.993	7.3605	+8.6	22.182 k	22.117 k	+0.29	74.3 min	6.14 s
45	Rh	[Kr] $4d^8\,5s^1$	8.087	7.4589	+8.4	23.289 k	23.220 k	+0.30	77.8 min	6.32 s
46	Pd	[Kr] $4d^{10}$	9.148	8.3368	+9.7	24.422 k	24.350 k	+0.30	81.5 min	29.14 s
47	Ag	[Kr] $4d^{10}\,5s^1$	8.229	7.5762	+8.6	25.593 k	25.514 k	+0.31	85.2 min	6.58 s
48	Cd	[Kr] $4d^{10}\,5s^2$	9.416	8.9938	+4.7	26.794 k	26.711 k	+0.31	89.0 min	8.00 s
49	In	[Kr] $4d^{10}\,5s^2\,5p^1$	5.681	5.7864	−1.8	28.029 k	27.940 k	+0.32	93.0 min	4.75 s
50	Sn	[Kr] $4d^{10}\,5s^2\,5p^2$	7.391	7.3439	+0.6	29.294 k	29.200 k	+0.32	97.0 min	6.56 s
51	Sb	[Kr] $4d^{10}\,5s^2\,5p^3$	8.998	8.6084	+4.5	30.592 k	30.491 k	+0.33	101.1 min	8.45 s
52	Te	[Kr] $4d^{10}\,5s^2\,5p^4$	9.078	9.0098	+0.8	31.921 k	31.814 k	+0.34	105.3 min	10.32 s
53	I	[Kr] $4d^{10}\,5s^2\,5p^5$	10.857	10.4512	+3.9	33.283 k	33.169 k	+0.34	109.6 min	12.30 s
54	Xe	[Kr] $4d^{10}\,5s^2\,5p^6$	12.560	12.1298	+3.5	34.676 k	34.561 k	+0.33	114.0 min	14.38 s

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
55	Cs	[Xe] $6s^1$	4.129	3.8939	+6.0	36.109 k	35.985 k	+0.34	118.5 min	2.53 s
56	Ba	[Xe] $6s^2$	5.395	5.2117	+3.5	37.574 k	37.441 k	+0.36	2.05 h	3.76 s
57	La	[Xe] $5d^1\,6s^2$	5.838	5.5769	+4.7	39.071 k	38.925 k	+0.38	2.13 h	4.50 s
58	Ce	[Xe] $4f^1\,5d^1\,6s^2$	5.906	5.5386	+6.6	40.594 k	40.443 k	+0.37	2.21 h	4.61 s
59	Pr	[Xe] $4f^3\,6s^2$	5.645	5.4702†	+3.2	42.144 k	41.991 k	+0.36	2.29 h	4.16 s
60	Nd	[Xe] $4f^4\,6s^2$	5.698	5.5248	+3.1	43.734 k	43.569 k	+0.38	2.37 h	4.24 s
61	Pm	[Xe] $4f^5\,6s^2$	5.750	5.5819	+3.0	45.359 k	45.184 k	+0.39	2.46 h	4.32 s
62	Sm	[Xe] $4f^6\,6s^2$	5.802	5.6437	+2.8	47.019 k	46.834 k	+0.39	2.55 h	4.40 s
63	Eu	[Xe] $4f^7\,6s^2$	5.854	5.6704	+3.2	48.713 k	48.519 k	+0.40	2.63 h	4.47 s
64	Gd	[Xe] $4f^7\,5d^1\,6s^2$	6.289	6.1498	+2.3	50.451 k	50.239 k	+0.42	2.72 h	5.21 s
65	Tb	[Xe] $4f^9\,6s^2$	6.099	5.8638	+4.0	52.212 k	51.996 k	+0.41	2.82 h	4.69 s
66	Dy	[Xe] $4f^{10}\,6s^2$	6.201	5.9391	+4.4	54.016 k	53.789 k	+0.42	2.91 h	4.78 s
67	Ho	[Xe] $4f^{11}\,6s^2$	6.298	6.0215	+4.6	55.857 k	55.618 k	+0.43	3.01 h	4.87 s
68	Er	[Xe] $4f^{12}\,6s^2$	6.392	6.1077	+4.7	57.736 k	57.486 k	+0.43	3.10 h	4.96 s
69	Tm	[Xe] $4f^{13}\,6s^2$	6.484	6.1844	+4.9	59.653 k	59.390 k	+0.44	3.20 h	5.05 s
70	Yb	[Xe] $4f^{14}\,6s^2$	6.576	6.2542	+5.2	61.609 k	61.332 k	+0.45	3.31 h	5.15 s
71	Lu	[Xe] $4f^{14}\,5d^1\,6s^2$	5.502	5.4259	+1.4	63.611 k	63.314 k	+0.47	3.41 h	10.60 s
72	Hf	[Xe] $4f^{14}\,5d^2\,6s^2$	6.959	6.8251	+2.0	65.655 k	65.351 k	+0.47	3.52 h	14.02 s
73	Ta	[Xe] $4f^{14}\,5d^3\,6s^2$	7.788	7.5496	+3.2	67.740 k	67.416 k	+0.48	3.62 h	7.60 s
74	W	[Xe] $4f^{14}\,5d^4\,6s^2$	8.005	7.8640	+1.8	69.867 k	69.525 k	+0.49	3.73 h	8.21 s
75	Re	[Xe] $4f^{14}\,5d^5\,6s^2$	8.186	7.8335	+4.5	72.037 k	71.676 k	+0.50	3.84 h	8.78 s
76	Os	[Xe] $4f^{14}\,5d^6\,6s^2$	8.911	8.4382	+5.6	74.250 k	73.871 k	+0.51	3.96 h	9.53 s
77	Ir	[Xe] $4f^{14}\,5d^7\,6s^2$	9.493	8.9670	+5.9	76.507 k	76.111 k	+0.52	4.07 h	10.20 s
78	Pt	[Xe] $4f^{14}\,5d^9\,6s^1$	9.776	8.9588	+9.1	78.805 k	78.395 k	+0.52	4.19 h	10.04 s
79	Au	[Xe] $4f^{14}\,5d^{10}\,6s^1$	9.890	9.2256	+7.2	81.152 k	80.725 k	+0.53	4.31 h	10.36 s
80	Hg	[Xe] $4f^{14}\,5d^{10}\,6s^2$	10.865	10.4375	+4.1	83.549 k	83.102 k	+0.54	4.44 h	11.93 s
81	Tl	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^1$	6.176	6.1083	+1.1	85.995 k	85.530 k	+0.54	4.56 h	7.24 s
82	Pb	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^2$	7.599	7.4167	+2.5	88.490 k	88.005 k	+0.55	4.69 h	9.33 s
83	Bi	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^3$	7.103	7.2855	−2.5	91.035 k	90.524 k	+0.56	4.82 h	8.49 s
84	Po	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^4$	8.377	8.4181	−0.5	93.630 k	93.105 k	+0.56	4.95 h	10.49 s
85	At	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^5$	9.617	9.3175	+3.2	96.277 k	95.730 k	+0.57	5.08 h	12.55 s
86	Rn	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^6$	10.841	10.7485	+0.9	98.974 k	98.404 k	+0.58	5.22 h	14.67 s
87	Fr	[Rn] $7s^1$	4.305	4.0727	+5.7	101.730 k	101.137 k	+0.59	5.36 h	3.23 s
88	Ra	[Rn] $7s^2$	5.496	5.2784	+4.1	104.541 k	103.922 k	+0.60	5.50 h	4.59 s
89	Ac	[Rn] $6d^1\,7s^2$	5.307	5.3802	−1.4	107.408 k	106.755 k	+0.61	5.65 h	8.20 s
90	Th	[Rn] $6d^2\,7s^2$	6.539	6.3067	+3.7	110.330 k	109.651 k	+0.62	5.80 h	6.53 s
91	Pa	[Rn] $5f^2\,6d^1\,7s^2$	5.741	5.8900†	−2.5	113.302 k	112.601 k	+0.62	5.95 h	9.67 s
92	U	[Rn] $5f^3\,6d^1\,7s^2$	5.820	6.1940	−6.0	116.335 k	115.606 k	+0.63	6.10 h	10.10 s
93	Np	[Rn] $5f^4\,6d^1\,7s^2$	5.856	6.2656	−6.5	119.432 k	–	–	6.26 h	10.43 s
94	Pu	[Rn] $5f^6\,7s^2$	6.108	6.0258	+1.4	122.582 k	–	–	6.42 h	5.84 s
95	Am	[Rn] $5f^7\,7s^2$	6.154	5.9738	+3.0	125.805 k	–	–	6.58 h	5.93 s
96	Cm	[Rn] $5f^7\,6d^1\,7s^2$	5.806	5.9922	−3.1	129.095 k	–	–	6.74 h	11.00 s
97	Bk	[Rn] $5f^9\,7s^2$	6.454	6.1978	+4.1	132.445 k	–	–	6.91 h	6.21 s
98	Cf	[Rn] $5f^{10}\,7s^2$	6.573	6.2819	+4.6	135.866 k	–	–	7.08 h	6.32 s
99	Es	[Rn] $5f^{11}\,7s^2$	6.683	6.3684	+4.9	139.360 k	–	–	7.26 h	6.43 s
100	Fm	[Rn] $5f^{12}\,7s^2$	6.788	6.5000†	+4.4	142.923 k	–	–	7.44 h	6.54 s
101	Md	[Rn] $5f^{13}\,7s^2$	6.890	6.5800†	+4.7	146.564 k	–	–	7.62 h	6.65 s
102	No	[Rn] $5f^{14}\,7s^2$	6.991	6.6262	+5.5	150.282 k	–	–	7.80 h	6.76 s
103	Lr	[Rn] $5f^{14}\,7s^2\,7p^1$	4.844	4.9600	−2.3	154.082 k	–	–	7.99 h	4.78 s
104	Rf	[Rn] $5f^{14}\,6d^2\,7s^2$ *	6.089	6.0200†	+1.2	157.965 k	–	–	8.18 h	12.72 s
105	Db	[Rn] $5f^{14}\,6d^3\,7s^2$ *	7.300	6.8000†	+7.4	161.934 k	–	–	8.38 h	15.74 s
106	Sg	[Rn] $5f^{14}\,6d^4\,7s^2$ *	8.459	7.8000†	+8.4	165.991 k	–	–	8.58 h	18.66 s
107	Bh	[Rn] $5f^{14}\,6d^5\,7s^2$ *	9.476	7.7000†	+23.1	170.138 k	–	–	8.78 h	12.45 s
108	Hs	[Rn] $5f^{14}\,6d^6\,7s^2$ *	8.508	7.6000†	+11.9	174.381 k	–	–	8.99 h	24.23 s
109	Mt	[Rn] $5f^{14}\,6d^7\,7s^2$ *	9.797	8.2914†	+18.2	178.705 k	–	–	9.20 h	26.99 s

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
110	Ds	[Rn] $5f^{14}\,6d^{8}\,7s^{2}$ *	11.029	9.9497†	+10.8	183.137 k	–	–	9.42 h	29.82 s
111	Rg	[Rn] $5f^{14}\,6d^{9}\,7s^{2}$ *	12.225	10.5716†	+15.6	187.674 k	–	–	9.64 h	32.70 s
112	Cn	[Rn] $5f^{14}\,6d^{10}\,7s^{2}$ *	13.396	11.9707†	+11.9	192.313 k	–	–	9.87 h	35.62 s
113	Nh	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{1}$ *	7.603	7.3058†	+4.1	197.070 k	–	–	10.10 h	11.76 s
114	Fl	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{2}$ *	8.895	8.6251†	+3.1	201.936 k	–	–	10.33 h	14.07 s
115	Mc	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{3}$ *	5.573	5.5791†	−0.1	206.927 k	–	–	10.57 h	7.27 s
116	Lv	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{4}$ *	6.831	6.8808†	−0.7	212.036 k	–	–	10.81 h	9.26 s
117	Ts	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{5}$ *	8.001	7.6996†	+3.9	217.278 k	–	–	11.06 h	11.26 s
118	Og	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{6}$ *	9.127	8.9133†	+2.4	222.656 k	–	–	11.32 h	13.30 s

D Alpha emitters

Gamow half-lives use $r_0 = 1.2$ fm and no fitted constant. Measured values marked † are alpha partial half-lives.

nuclide	Q (MeV)	β	η	$2G$	Gamow $t_{1/2}$	measured	$\log_{10}$ ratio
Te-106	4.290	0.0489	14.92	39.5	97.7 µs	78 µs	+0.10
Nd-144	1.901	0.0324	26.13	101.6	4.74×10^{15} yr	2.29×10^{15} yr	+0.32
Sm-146	2.529	0.0374	23.44	83.8	7.83×10^{7} yr	9.2×10^{7} yr	−0.07
Sm-148	1.987	0.0331	26.45	102.3	9.65×10^{15} yr	6.3×10^{15} yr	+0.19
Gd-148	3.271	0.0425	21.30	69.5	43.6 yr	86.9 yr	−0.30
Dy-154	2.945	0.0403	23.19	79.7	1.29×10^{6} yr	1.4×10^{6} yr	−0.03
Pt-190	3.269	0.0423	26.21	90.5	6.32×10^{10} yr	4.83×10^{11} yr	−0.88
Po-210	5.408	0.0544	22.00	61.2	3.4 d	138 d	−1.61
Po-212	8.954	0.0700	17.10	32.3	0.0629 µs	0.294 µs	−0.67
Po-214	7.834	0.0654	18.29	39.0	55.4 µs	163 µs	−0.47
Po-216	6.906	0.0614	19.48	45.9	56.8 ms	144 ms	−0.40
Po-218	6.115	0.0578	20.70	53.0	77.6 s	3.1 min	−0.38
Rn-220	6.405	0.0592	20.72	52.2	33.6 s	55.6 s	−0.22
Rn-222	5.590	0.0553	22.18	60.8	2.33 d	3.82 d	−0.21
Ra-224	5.789	0.0562	22.32	60.7	2.02 d	3.63 d	−0.25
Ra-226	4.871	0.0516	24.33	72.8	1.05×10^{3} yr	1.6×10^{3} yr	−0.18
Th-228	5.520	0.0549	23.39	66.1	1.27 yr	1.91 yr	−0.18
Th-230	4.770	0.0510	25.16	76.8	5.78×10^{4} yr	7.54×10^{4} yr	−0.12
Th-232	4.082	0.0472	27.21	89.1	1.47×10^{10} yr	1.4×10^{10} yr	+0.02
U-234	4.857	0.0515	25.51	77.8	1.62×10^{5} yr	2.46×10^{5} yr	−0.18
U-236	4.573	0.0500	26.29	82.5	1.79×10^{7} yr	2.34×10^{7} yr	−0.12
U-238	4.271	0.0483	27.21	88.0	4.58×10^{9} yr	4.46×10^{9} yr	+0.01
Pu-238	5.593	0.0552	24.30	69.5	36.3 yr	87.7 yr	−0.38
Pu-240	5.256	0.0536	25.07	74.0	3.59×10^{3} yr	6.56×10^{3} yr	−0.26
Pu-242	4.984	0.0521	25.75	78.0	2.03×10^{5} yr	3.75×10^{5} yr	−0.27
Pu-244	4.666	0.0504	26.61	83.2	3.74×10^{7} yr	8.14×10^{7} yr†	−0.34
Cm-244	5.902	0.0567	24.18	67.6	5.54 yr	18.1 yr	−0.51
Cm-248	5.162	0.0531	25.86	77.6	1.26×10^{5} yr	3.8×10^{5} yr†	−0.48
Cf-252	6.217	0.0582	24.07	65.7	307 d	2.73 yr†	−0.51
Fm-254	7.307	0.0631	22.66	56.5	38.1 min	3.24 h	−0.71
No-254	8.226	0.0670	21.79	50.6	5.89 s	56.9 s†	−0.99
Hs-270	9.070	0.0703	22.01	48.8	1.04 s	7.6 s	−0.86
Fl-288	10.076	0.0740	22.08	46.3	75.8 ms	653 ms	−0.94
Lv-292	10.791	0.0766	21.71	43.3	3.81 ms	16 ms	−0.62
Og-294	11.867	0.0803	21.07	38.9	42.6 µs	580 µs	−1.13

E The radioactive edge: longest-lived isotopes

NUBASE2020 values as quoted on the isotope pages. Bracketed values are alternatives printed there.

Z	isotope	half-life	mode
43	Tc-97	$4.21(16)\times10^{6}$ y	EC
61	Pm-145	17.7(4) y	EC
83	Bi-209	$2.01(8)\times10^{19}$ y	α
84	Po-209	124(3) y	α
85	At-210	8.1(4) h	EC/β^+
86	Rn-222	3.8215(2) d	α
87	Fr-223	22.00(7) min	β^-
88	Ra-226	1600(7) y	α
89	Ac-227	21.772(3) y	β^-
90	Th-232	$1.40(1)\times10^{10}$ y	α
91	Pa-231	$3.265(20)\times10^{4}$ y	α
92	U-238	$4.463(3)\times10^{9}$ y	α
93	Np-237	$2.144(7)\times10^{6}$ y	α
94	Pu-244	$8.13(3)\times10^{7}$ y	α
95	Am-243	7350(9) y	α
96	Cm-247	$1.56(5)\times10^{7}$ y	α
97	Bk-247	$1.38(25)\times10^{3}$ y	α
98	Cf-251	898(44) y	α
99	Es-252	471.7(19) d	α
100	Fm-257	100.5(2) d	α
101	Md-258	51.59(29) d	α
102	No-259	58(5) min	α
103	Lr-266	11 ($+21/-5$) h [22(14) h]	SF
104	Rf-267	48 ($+23/-12$) min	SF
105	Db-268	16 ($+6/-4$) h	α/SF
106	Sg-269	13 ($+6/-4$) min	α
107	Bh-270	2.4 ($+4.4/-0.9$) min	α
108	Hs-271	46 ($+56/-16$) s	α
109	Mt-278	4.5 ($+3.5/-1.3$) s [6(3) s]	α
110	Ds-281	14(3) s	SF
111	Rg-282	100 ($+70/-30$) s	α
112	Cn-285	28 ($+9/-6$) s [30(8) s]	α
113	Nh-286	9.5 ($+6.3/-2.7$) s [12(5) s]	α
114	Fl-289	1.9 ($+0.7/-0.4$) s [2.1(6) s]	α
115	Mc-290	650 ($+490/-200$) ms	α
116	Lv-293	57 ($+43/-17$) ms [70(30) ms]	α
117	Ts-294	51 ($+38/-16$) ms [70(30) ms]	α
118	Og-294	0.58 ($+0.44/-0.18$) ms [0.7(3) ms]	α

References

[1] M. E. Rose and T. A. Welton, "The virial theorem for a Dirac particle," Phys. Rev. **86**, 432 (1952).

[2] A. Sommerfeld, Ann. Phys. (Leipzig) **51**, 1 (1916).

[3] W. Pieper and W. Greiner, Z. Phys. **218**, 327 (1969).

[4] W. Kohn and L. J. Sham, Phys. Rev. **140**, A1133 (1965).

[5] J. P. Perdew and A. Zunger, Phys. Rev. B **23**, 5048 (1981).

[6] J. C. Slater, Adv. Quantum Chem. **6**, 1 (1972).

[7] J. F. Janak, Phys. Rev. B **18**, 7165 (1978).

[8] G. Gamow, Z. Phys. **51**, 204 (1928).

[9] R. W. Gurney and E. U. Condon, Nature **122**, 439 (1928).

[10] M. Weitz *et al.*, Phys. Rev. A **52**, 2664 (1995).

[11] G. Herzberg, "Lamb shift of the 1^2S ground state of deuterium," Proc. R. Soc. Lond. A **234**, 516 (1956).

[12] N. Parker, "Revisiting the deuterium Lyman alpha line experiment," Physics Forums Insights (2019, revised 2024), https://www.physicsforums.com/insights/the-deuterium-lyman-alpha-line/.

[13] A. Gumberidze *et al.*, Phys. Rev. Lett. **94**, 223001 (2005).

[14] H. G. J. Moseley, Phil. Mag. **26**, 1024 (1913).

[15] A. C. Thompson *et al.*, *X-ray Data Booklet*, Lawrence Berkeley National Laboratory, Tables 1-1 and 1-2, https://xdb.lbl.gov.

[16] J. E. Sansonetti and W. C. Martin, "Handbook of basic atomic spectroscopic data," J. Phys. Chem. Ref. Data **34**, 1559 (2005); https://physics.nist.gov/PhysRefData/Handbook.

[17] F. G. Kondev *et al.*, "The NUBASE2020 evaluation of nuclear physics properties," Chin. Phys. C **45**, 030001 (2021).

[18] M. Wang *et al.*, "The AME 2020 atomic mass evaluation (II). Tables, graphs and references," Chin. Phys. C **45**, 030003 (2021).

[19] U. Volz and H. Schmoranzer, Phys. Scr. **T65**, 48 (1996).

[20] R. Lange *et al.*, Phys. Rev. Lett. **126**, 011102 (2021).

[21] A. Kramida, Yu. Ralchenko, J. Reader and NIST ASD Team, *NIST Atomic Spectra Database*, https://physics.nist.gov/asd.